Biostatistical Methods I Practice Problems

1 Statistics and Delta Method

1. For $i = 1, \ldots, n$, let X_i be iid random variables with $E(X_i) = \mu$ and $Var(X_i) = \sigma^2$. Approximate the distribution of $1/\bar{X}$ with $\bar{X} = \frac{1}{n}\sum X_i$.

2. In a linear regression,

$$(Y_i|X_i = x_i) \sim N(\beta_0 + \beta_1 x_i, \sigma^2), \quad i = 1, \ldots, n$$

Show that the sufficient statistics for $(\beta_0, \beta_1, \sigma^2)$ are $(\hat{\beta}_0, \hat{\beta}_1, \hat{\sigma^2}, \sum X_i, \sum X_i^2)$ with

$$\begin{aligned}
\hat{\beta}_0 &= \bar{y} - \hat{\beta}_1 \bar{x} \\
\hat{\beta}_1 &= \sum (x_i - \bar{x})(y_i - \bar{y}) / \sum (x_i - \bar{x})^2 \\
\hat{\sigma^2} &= \sum (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2 / (n-2).
\end{aligned}$$

2 Likelihood and Statistical Tests

1. Consider a single observation from a Poisson distribution: $Y|\theta \sim \text{Poisson}(\theta)$.

   (a) Suppose we wish to estimate $\exp(-3\theta)$. Show that the UMVUE is $(-2)^y$ for $y = 0, 1, 2, \ldots$ Is this a reasonable estimator?

   (b) Suppose we wish to estimate θ^2. Show that $Y(Y-1)$ is the UMVUE for θ^2 for $Y = 1$. Comment on whether this is a sensible estimator.

For questions 2 and 3, suppose that we have data Y_i independently following a Poisson distribution with $E(Y_i) = \lambda$, for $i = 1, \ldots, n$.

2. The investigator is interested in testing $H_0 : \lambda = \lambda_0$.

   (a) Derive the score test statistic and its approximate null distribution.

   (b) Derive the Wald test statistic and its approximate null distribution.

   (c) Derive the likelihood ratio test statistic and its approximate null distribution

3. Let $\theta = \log(\lambda)$, the null becomes $H_0 : \theta = \theta_0$ with $\theta_0 = \log(\lambda_0)$.

   (a) Show that the score test statistic is invariant to parametrization.

   (b) Show that the Wald test statistic is not invariant to parametrization.

   (c) Show that the likelihood ratio test statistic is invariant to parametrization.

4. Suppose that the sample mean was 0.6 from $n = 20$ trials and $\lambda_0 = 1$, compute the Wald statistic and perform the Wald test for testing the null hypothesis $H_0 : \lambda = \lambda_0$ and $H_0 : \theta = \theta_0$, separately. Compare the results and the conclusions you can make.

5. Suppose $Y_i|\theta_i \underset{iid}{\sim} \text{Poisson}(\theta_i)$ with $\theta_i \underset{iid}{\sim} \text{Ga}(\mu_i b, b)$ for $i = 1, \ldots, n$.

   (a) Show that $E[Y_i] = \mu_i$ and $\text{var}(Y_i) = \mu_i(1 + b^{-1})$.

   (b) Show that the marginal distribution of $Y_i|\mu_i, b$ is negative binomial.

3 Bayesian Statistics and Priors

1. Derive the posterior mean and posterior quantiles as the solution to quadratic and linear loss.

2. Consider the Neyman-Scott problem in which $Y_{ij}|\mu_i, \sigma^2 \underset{ind}{\sim} N(\mu_i, \sigma^2),\ i =, 1 \ldots, n,\ j = 1, 2.$

   (a) Show that Jeffreys prior in this case is

   $$\pi(\mu_1, \ldots, \mu_n, \sigma^2) \propto \sigma^{-n-2}$$

   (b) Derive the posterior distribution corresponding to this prior and show that

   $$E[\sigma^2|y] = \frac{1}{2(n-1)} \sum_{i=1}^{n} \frac{(Y_{i1} - Y_{i2})^2}{2}.$$

   (c) Hence, using exercise 2.6 (below)

   > **2.6.** Consider the so-called Neyman-Scott problem in which
   >
   > $$Y_{ij}|\mu_i, \sigma^2 \underset{ind}{\sim} N(\mu_i, \sigma^2),\ i =, 1 \ldots, n,\ j = 1, 2.$$
   >
   > Obtain the MLE of σ^2 and show that it is inconsistent. Why does the inconsistency rise in this example.

   show that $E[\sigma^2|y] \to \sigma^2/2$ as $n \to \infty$, so that the posterior mean is inconsistent.

   (d) Examine the posterior distribution corresponding to the prior

   $$\pi(\mu_1, \ldots, \mu_n, \sigma^2) \propto \sigma^{-2}$$

   (e) Is the posterior mean for σ^2 consistent in this case?

3. Consider the "likelihood", $\hat{\theta}|\theta \sim N(\theta, V)$ and the prior $\theta \sim N(0, W)$ with V and W known. Show that $\theta|\hat{\theta} \sim N(r\hat{\theta}, rV)$ where $r = W/(V + W)$.

4. Exercise 3.4 in Jon Wakefield's *Bayesian and Frequentist Regression Methods* - too lazy to type, here is a screenshot

3.4 Consider the data given in Table 3.4, which are a simplified version of those reported in Breslow and Day (1980). These data arose from a case-control study (Sect. 7.10) that was carried out to investigate the relationship between esophageal cancer and various risk factors. There are 200 cases and 775 controls. Disease status is denoted Y with $Y = 0/1$ corresponding to without/with disease, and alcohol consumption is represented by X with $X = 0/1$ denoting $< 80\,\mathrm{g}/ \geq 80\,\mathrm{g}$ on average per day. Let the probabilities of high alcohol consumption in the cases and controls be denoted

$$p_1 = \Pr(X = 1 \mid Y = 1) \quad \text{and} \quad p_2 = \Pr(X = 1 \mid Y = 0),$$

respectively. Further, let X_1 be the number exposed from n_1 cases and X_2 be the number exposed from n_2 controls. Suppose $X_i \mid p_i \sim \text{Binomial}(n_i, p_i)$ in the case ($i = 1$) and control ($i = 2$) groups.

(a) Of particular interest in studies such as this is the *odds ratio* defined by

$$\theta = \frac{\Pr(Y = 1 \mid X = 1)/\Pr(Y = 0 \mid X = 1)}{\Pr(Y = 1 \mid X = 0)/\Pr(Y = 0 \mid X = 0)}.$$

Show that the odds ratio is equal to

$$\theta = \frac{\Pr(X = 1 \mid Y = 1)/\Pr(X = 0 \mid Y = 1)}{\Pr(X = 1 \mid Y = 0)/\Pr(X = 0 \mid Y = 0)} = \frac{p_1/(1-p_1)}{p_2/(1-p_2)}.$$

(b) Obtain the MLE and a 90% confidence interval for θ, for the data of Table 3.4.

(c) We now consider a Bayesian analysis. Assume that the prior distribution for p_i is the beta distribution $\text{Be}(a, b)$ for $i = 1, 2$. Show that the posterior distribution $p_i \mid x_i$ is given by the beta distribution $\text{Be}(a+x_i, b+n_i-x_i)$, $i = 1, 2$.

(d) Consider the case $a = b = 1$. Obtain expressions for the posterior mean, mode, and standard deviation. Evaluate these posterior summaries for the data of Table 3.4. Report 90% posterior credible intervals for p_1 and p_2.

(e) Obtain the asymptotic form of the posterior distribution and obtain 90% credible intervals for p_1 and p_2. Compare this interval with the exact calculation of the previous part.

(f) Simulate samples $p_1^{(t)}, p_2^{(t)}$, $t = 1, \ldots, T = 1{,}000$ from the posterior distributions $p_1 \mid x_1$ and $p_2 \mid x_2$. Form histogram representations of the posterior distributions using these samples, and obtain sample-based 90% credible intervals.

(g) Obtain samples from the posterior distribution of $\theta \mid x_1, x_2$ and provide a histogram representation of the posterior. Obtain the posterior median and 90% credible interval for $\theta \mid x_1, x_2$ and compare with the likelihood analysis.

(h) Suppose the rate of esophageal cancer is 17 in 100,000. Describe how this information may be used to evaluate

$$q_1 = \Pr(Y = 1 \mid X = 1) \quad \text{and} \quad q_0 = \Pr(Y = 1 \mid X = 0).$$

4 Modeling and Statistical Testing

1. For the linear model $y = x\beta + \epsilon$ with $\epsilon \sim N(0, \sigma^2 \mathbf{I}_n)$, show that the joint confidence interval for $\beta(p \times 1)$ is the region for the β such that $(\beta - \hat{\beta})^T (X^T X)(\beta - \hat{\beta}) \leq p\hat{\sigma}^2 F_{p,m-p}^{(1-\alpha)}$, with $\hat{\beta} = (X^T X)^{-1} X^T y$ and $\hat{\sigma}^2 = (y - X\hat{\beta})^T (y - X\hat{\beta})/(n-p)$.

2. A pharmacokinetic experiment was conducted to measure the concentration of the drug Carvedilol, a beta blocker, after a single dose of 30mg was administered to a cardiac failure patient. The response y_i represents the drug concentration (in mg/L) at time x_i (in hours) for $i = 1, \ldots, 8$.

i	x_i	y_i
1	2	1.63
2	4	1.01
3	6	0.73
4	8	0.55
5	10	0.41
6	24	0.01
7	28	0.06
8	32	0.02

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```
 i = c(1:8) #index
 xi = c(2, 4, 6, 8, 10, 24, 28, 32) # hours
yi = c(1.63, 1.01, 0.73, 0.55, 0.41, 0.01, 0.06, 0.02) # mg/L
```

The following model was considered

$$\log(y_i) = \mu_i(\beta) + \epsilon_i = \log\left[\frac{D}{V}\exp(-k_e x_i)\right] + \epsilon_i,$$

with $D = 30$ and $\epsilon_i \overset{iid}{\sim} N(0, \sigma^2$. The parameters $\beta = [V, k_e]$ are the volume of the distribution $V > 0$ and the elimination rate k_e.

(a) For this model, obtain expressions for,

   i. The log-likelihood function $L(\beta, \sigma^2)$.

   ii. The score function $S(\beta, \sigma^2)$.

   iii. The expected information matrix $I(\beta, \sigma^2)$.

(b) Obtain the MLEs for β and σ^2 and give an asymptotic 95% CI for each parameter. Describe your approach.

(c) The clearance $Cl = V \times k_\epsilon$ and elimination half-life $x_{1/2} = \log(2)/k_\epsilon$ are parameters of interest in this experiment. Find the MLEs of these parameters along with asymptotic 95% CI's.

(d) Examine the fit of the model by plotting the observed data along with the fitted curve. Comment on the fit.

(e) Use residuals to examine the appropriateness of the assumptions of the model. Do the assumptions seem reasonable for these data?

(f) Describe any potential pitfalls of the model assumptions.

5 Fall 2023 Exam 1

Suppose $X_1, \ldots, X_n$ are i.i.d. such that $X_i \sim \text{NegBin}(\mu, b)$ for all $i = 1, \ldots, n$. The probability mass function for any given X is

$$P[X = x|\mu, b] = \frac{\Gamma(x+b)}{\Gamma(x+1)\Gamma(b)} \left(\frac{\mu}{\mu+b}\right)^x \left(\frac{b}{\mu+b}\right)^b, \quad x = 0, 1, 2, \ldots$$

where $b > 0$ is known and $\mu > 0$ is an unknown parameter. This implies that $E[X] = \mu$ and $\text{var}[X] = \mu + \mu^2/b$. Based on these n i.i.d. outcomes:

1. What is the likelihood function of μ?
   The likelihood function of X can be represented as the joint probability of all X's.

$$\begin{aligned} L_n(\mu|b, X) &= \prod_{i=1}^n P[X_i = x_i|\mu, b] \\ &= \prod_{i=1}^n \left[\frac{\Gamma(x_i+b)}{\Gamma(x_i+1)\Gamma(b)} \left(\frac{\mu}{\mu+b}\right)^{x_i} \left(\frac{b}{\mu+b}\right)^b\right] \\ &= \left[\left(\frac{b}{\mu+b}\right)^{nb}\right] \prod_{i=1}^n \left[\frac{\Gamma(x_i+b)}{\Gamma(x_i+1)\Gamma(b)} \left(\frac{\mu}{\mu+b}\right)^{x_i}\right] \end{aligned}$$

   The log-likelihood follows as

$$\begin{aligned} l_n(\mu|b, X) &= \left[nb \log\left(\frac{b}{\mu+b}\right)\right] + \sum_{i=1}^n \log\left[\frac{\Gamma(x_i+b)}{\Gamma(x_i+1)\Gamma(b)} \left(\frac{\mu}{\mu+b}\right)^{x_i}\right] \\ &= \left[nb \log\left(\frac{b}{\mu+b}\right)\right] + \sum_{i=1}^n \log\left[\frac{\Gamma(x_i+b)}{\Gamma(x_i+1)\Gamma(b)}\right] + \log\left(\frac{\mu}{\mu+b}\right) \sum_{i=1}^n x_i \end{aligned}$$

2. Derive the score function of μ.
   The score function is the partial derivative of the log-likelihood function in relation to μ.

$$\begin{aligned} S_n(\mu) &= \frac{\partial}{\partial\mu} l(\mu|b, X) \\ &= \frac{\partial}{\partial\mu}\left[\left[nb \log\left(\frac{b}{\mu+b}\right)\right] + \log\left(\frac{\mu}{\mu+b}\right) \sum_{i=1}^n x_i\right] \\ &= nb \cdot \frac{\partial}{\partial\mu} \log\left(\frac{b}{\mu+b}\right) + \left[\sum_{i=1}^n x_i\right] \cdot \frac{\partial}{\partial\mu} \log\left(\frac{\mu}{\mu+b}\right) \end{aligned}$$

by the chain and quotient rule

$$
\begin{aligned}
&= nb \cdot \left[\frac{\mu+b}{b}\right]\left[\frac{(\mu+b)(0)-(b)(1)}{(\mu+b)^2}\right] + \left[\sum_{i=1}^{n} x_i\right] \cdot \left[\frac{\mu+b}{\mu}\right]\left[\frac{(\mu+b)(1)-(\mu)(1)}{(\mu+b)^2}\right] \\
&= \left[\frac{-nb}{\mu+b}\right] + \left[\sum_{i=1}^{n} x_i\right] \cdot \left[\frac{b}{\mu(\mu+b)}\right] \\
S_n(\mu) &= \frac{b}{\mu+b}\left[\frac{1}{\mu}\sum_{i=1}^{n} x_i - n\right]
\end{aligned}
$$

3. Show that Fisher's expected information in a sample of size n is

$$
I_n(\mu) = \frac{nb}{\mu(\mu+b)}
$$

Fisher's expected information for μ is the negative expected value of the partial derivative of the score function of μ with respect to μ.

$$
\begin{aligned}
I_n(\mu) &= -E\left[\frac{\partial}{\partial\mu} S_n(\mu)\right] \\
&= -E\left[\frac{\partial}{\partial\mu}\left(\frac{b}{\mu+b}\left[\frac{1}{\mu}\sum_{i=1}^{n} x_i - n\right]\right)\right] \\
&= -E\left[\left(-\frac{b}{(\mu+b)^2}\right)\left[\frac{1}{\mu}\sum_{i=1}^{n} x_i - n\right] + \left[\frac{-1}{\mu^2}\sum_{i=1}^{n} x_i\right]\left(\frac{b}{\mu+b}\right)\right] \\
&= -\left[\left(-\frac{b}{(\mu+b)^2}\right)\left[\frac{1}{\mu}(n\mu) - n\right]\right] - \left[\left[\frac{-1}{\mu^2}(n\mu)\right]\left(\frac{b}{\mu+b}\right)\right] \\
&= -\left[\left[\frac{-1}{\mu^2}(n\mu)\right]\left(\frac{b}{\mu+b}\right)\right] \\
&= \frac{nb}{\mu(\mu+b)}
\end{aligned}
$$

4. Find the MLE of $\hat{\mu}$.
The MLE of $\hat{\mu}$ is the solution to the score function of μ when set equal to zero.

$$
\begin{aligned}
S_n(\hat{\mu}) = 0 &= \frac{b}{\hat{\mu}+b}\left[\frac{1}{\hat{\mu}}\sum_{i=1}^{n} x_i - n\right] \\
0 &= \frac{1}{\hat{\mu}}\sum_{i=1}^{n} x_i - n \\
\hat{\mu} &= \frac{1}{n}\sum_{i=1}^{n} x_i = \bar{x}
\end{aligned}
$$

5. What is the model based variance estimator for $\hat{\mu}$?
The model based variance estimator for μ is derived from the asymptotic distribution of $\hat{\mu}$. From sandwich estimation we can see that the estimated variance of $\hat{\mu}$ is

$$\sqrt{n}(\hat{\mu}-\mu) \xrightarrow{d} N(0, I_1^{-1}(\mu)) \implies \hat{V}(\hat{\mu}) = I_1^{-1}(\hat{\mu}) = \left[\frac{1}{n}\frac{nb}{\hat{\mu}(\hat{\mu}+b)}\right]^{-1} = \hat{\mu}+\hat{\mu}^2/b$$

6. Specify an asymptotic 95% confidence interval for μ.
The 95% confidence interval for μ can be derived easily from the asymptotic equation above. Let $Z \sim N(0,1)$ and $Z_\alpha = P(Z < \alpha)$, then if we let $\alpha = 0.05$ the following is the 95% confidence interval for μ:

$$\mu \in \left(\hat{\mu} \pm Z_{\alpha/2} \cdot \sqrt{\hat{\mu}+\hat{\mu}^2/b}\right).$$

7. Suppose we wish to test $H_0 : \mu = \mu_0$ versus $H_1 : \mu \neq \mu_0$.

   (a) Derive the score test statistic and its asymptotic null distribution.
   The score test statistic is derived from the first term Taylor expansion of the score function of the statistic around the value of the estimated statistic.

$$\begin{aligned} 0 = S_n(\hat{\mu}) &= S_n(\mu) + (\hat{\mu}-\mu)\frac{\partial}{\partial\mu}S_n(\tilde{\mu}) \\ \sqrt{n}(\hat{\mu}-\mu) &= \frac{\frac{1}{\sqrt{n}}S_n(\mu)}{\frac{1}{n}\frac{\partial}{\partial\mu}S_n(\tilde{\mu})} \\ \implies \frac{\frac{1}{\sqrt{n}}S_n(\mu)}{\frac{1}{n}\frac{\partial}{\partial\mu}S_n(\tilde{\mu})} &\sim \frac{N(0, I_1(\mu))}{I_1(\mu)} = N(0, I_1^{-1}(\mu)). \end{aligned}$$

   This gives the test statistic

$$S_n(\mu)I_1^{-1}(\mu)S_n(\mu)/n \sim \chi_1^2$$

   The corresponding statistic for testing μ_0 is exactly,

$$\left(\frac{b}{\mu_0+b}\left[\frac{1}{\mu_0}\sum_{i=1}^n x_i - n\right]\right)\left(\frac{\mu_0(\mu_0+b)}{b}\right)\left(\frac{b}{\mu_0+b}\left[\frac{1}{\mu_0}\sum_{i=1}^n x_i - n\right]\right)/n \sim \chi_1^2$$

   which simplified becomes,

$$\frac{nb[\bar{x}-\mu_0]^2}{\mu_0(\mu_0+b)} \sim \chi_1^2$$

   (b) Derive the Wald test statistic and its asymptotic null distribution.
   Similarly, the Wald test statistic is derived from the first term Taylor expansion,

$$\begin{aligned} 0 = S(\hat{\mu}) &= S(\mu) + (\hat{\mu}-\mu)\frac{\partial}{\partial\mu}S(\tilde{\mu}) \\ \sqrt{n}(\hat{\mu}-\mu) &= \frac{\frac{1}{\sqrt{n}}S(\mu)}{\frac{1}{n}\frac{\partial}{\partial\mu}S(\tilde{\mu})} \\ \implies \sqrt{n}(\hat{\mu}-\mu) &\sim N(0, I_1^{-1}(\mu)). \end{aligned}$$

This gives the test statistic

$$n(\hat{\mu}-\mu)I_1(\mu)(\hat{\mu}-\mu) \sim \chi_1^2$$

The corresponding statistic for μ is exactly,

$$n(\bar{x}-\mu_0)\left(\frac{b}{\mu_0(\mu_0+b)}\right)(\bar{x}-\mu_0) \sim \chi_1^2$$

which simplified becomes,

$$\frac{nb[\bar{x}-\mu_0]^2}{\mu_0(\mu_0+b)} \sim \chi_1^2$$

(c) Derive the likelihood ratio test statistic and its asymptotic null distribution.
The likelihood ratio test statistic comes from the second term Taylor expansion.

$$l_n(\mu) = l_n(\hat{\mu}) + \underbrace{(\mu-\hat{\mu})\left[\frac{\partial}{\partial\mu}l_n(\mu)\right]_{\hat{\mu}=\mu}}_{\text{becomes 0}} + \underbrace{\frac{1}{2}(\mu-\hat{\mu})^T\left[\frac{\partial^2}{\partial\mu\partial\mu^T}l_n(\mu)\right]_{\hat{\mu}=\tilde{\mu}}(\mu-\hat{\mu})}_{\text{asymptotically equivalent to the final form of the Wald test.}}$$

$$2[l_n(\mu)-l_n(\hat{\mu})] \approx n(\mu-\hat{\mu})^T I_1(\mu)(\mu-\hat{\mu}) \xrightarrow{d} \chi_1^2$$

where $\tilde{\mu}$ is between μ and $\hat{\mu}$. This gives the test statistic

$$2[l_n(\mu)-l_n(\hat{\mu})] \sim \chi_1^2$$

The corresponding statistic for μ is exactly,

$$2\left[nb\log\left(\frac{b}{\mu_0+b}\right)+n\bar{x}\log\left(\frac{\mu_0}{\mu_0+b}\right)-nb\log\left(\frac{b}{\bar{x}+b}\right)-n\bar{x}\log\left(\frac{\bar{x}}{\bar{x}+b}\right)\right] \sim \chi_1^2$$

which simplified becomes,

$$2n\left[b+\bar{x}\right]\log\left(\frac{\bar{x}+b}{\mu_0+b}\right) \sim \chi_1^2$$

8. Let $p=\mu/(\mu+b)$ and $p_0=\mu_0/(\mu_0+b)$. Consider testing $H_0: p=p_0$ versus $H_1: p\neq p_0$. Show that the Wald test statistic is not invariant under re-parameterization compared to the corresponding statistic of Question 7.
We can reparameterize the log-likelihood function with p and derive all three tests from

there.

$$
\begin{aligned}
l_n(\mu|b,X) &= \left[nb\log\left(\frac{b}{\mu+b}\right)\right] + \log\left(\frac{\mu}{\mu+b}\right)\sum_{i=1}^{n} x_i + \sum_{i=1}^{n}\log\left[\frac{\Gamma(x_i+b)}{\Gamma(x_i+1)\Gamma(b)}\right] \\
l_n(p|b,X) &= nb\log(1-p) + n\bar{x}\log(p) + \sum_{i=1}^{n}\log\left[\frac{\Gamma(x_i+b)}{\Gamma(x_i+1)\Gamma(b)}\right] \\
S_n(p) &= \frac{n\bar{x}}{p} - \frac{nb}{1-p} = \frac{n\bar{x}(1-p)-nbp}{p(1-p)} \\
I_n(p) &= -E\left(\frac{\partial}{\partial p}\left[\frac{n\bar{x}}{p} - \frac{nb}{1-p}\right]\right) \\
&= \frac{n\mu}{p^2} + \frac{nb}{(1-p)^2} \\
I_1(p) &= \frac{\mu}{p^2} + \frac{b}{(1-p)^2} = \frac{\mu(1-p)^2+bp^2}{p^2(1-p)^2} = \frac{b}{p(1-p)^2}
\end{aligned}
$$

The score test statistic is then,

$$
\begin{aligned}
&\left[\frac{n\bar{x}(1-p_0)-nbp_0}{p_0(1-p_0)}\right]\left[\frac{p_0(1-p_0)^2}{b}\right]\left[\frac{n\bar{x}(1-p_0)-nbp_0}{p_0(1-p_0)}\right]/n \\
=&\left[\frac{n\bar{x}\left(\frac{b}{\mu_0+b}\right)-nb\left(\frac{\mu_0}{\mu_0+b}\right)}{\left(\frac{\mu_0}{\mu_0+b}\right)\left(\frac{b}{\mu_0+b}\right)}\right]\left[\frac{\left(\frac{\mu_0}{\mu_0+b}\right)\left(\frac{b}{\mu_0+b}\right)^2}{b}\right]\left[\frac{n\bar{x}\left(\frac{b}{\mu_0+b}\right)-nb\left(\frac{\mu_0}{\mu_0+b}\right)}{\left(\frac{\mu_0}{\mu_0+b}\right)\left(\frac{b}{\mu_0+b}\right)}\right]/n \\
=&\left[\frac{1}{nb\left(\frac{\mu_0}{\mu_0+b}\right)}\right]\left[n\bar{x}\left(\frac{b}{\mu_0+b}\right)-nb\left(\frac{\mu_0}{\mu_0+b}\right)\right]^2 \\
=&\left[\frac{\mu_0+b}{nb\mu_0}\right]\left[\frac{nb\bar{x}}{\mu_0+b}-\frac{nb\mu_0}{\mu_0+b}\right]^2 \\
=&\left[\frac{n^2b^2(\mu_0+b)}{nb\mu_0(\mu_0+b)^2}\right][\bar{x}-\mu_0]^2 \\
=&\frac{nb(\mu_0-\bar{x})^2}{\mu_0(\mu_0+b)}
\end{aligned}
$$

which shows it is invariant to re-parameterization.
The likelihood ratio test statistic is

$$
\begin{aligned}
&2\left[nb\log(1-p_0)+n\bar{x}\log(p_0)-nb\log(1-p)-n\bar{x}\log(p)\right] \\
=&2\left[nb\log\left(\frac{b}{\mu_0+b}\right)+n\bar{x}\log\left(\frac{\mu_0}{\mu_0+b}\right)-nb\log\left(\frac{b}{\mu+b}\right)-n\bar{x}\log\left(\frac{\mu}{\mu+b}\right)\right] \\
=&2\left[nb\log\left(\frac{\mu+b}{\mu_0+b}\right)+n\bar{x}\log\left(\frac{\mu+b}{\mu_0+b}\right)\right] \\
=&2n[b+\bar{x}]\log\left(\frac{\mu+b}{\mu_0+b}\right)
\end{aligned}
$$

which can be seen as invariant when used for estimation and μ is estimated by its MLE $\bar{x}$. The Wald test statistic is

$$
\begin{aligned}
& n(p-p_0)\left(\frac{b}{p_0(1-p_0)^2}\right)(p-p_0) \\
=& n\left(\left(\frac{\mu}{\mu+b}\right)-\left(\frac{\mu_0}{\mu_0+b}\right)\right)\left(\frac{b}{\left(\frac{\mu_0}{\mu_0+b}\right)\left(\frac{b}{\mu_0+b}\right)^2}\right)\left(\left(\frac{\mu}{\mu+b}\right)-\left(\frac{\mu_0}{\mu_0+b}\right)\right) \\
=& \frac{nb(\mu_0-\mu)^2(\mu_0+b)}{\mu_0(\mu+b)^2}
\end{aligned}
$$

which clearly does not simply to what was originally found without re-parameterization because the term $(\mu+b)$ in the denominator is not based on μ_0, if it were then it would be invariant, but it is not.

6 Fall 2023 Exam 2

1. Suppose $X_1, \ldots, X_n$ are i.i.d. Poisson(μ) for an unknown $\mu > 0$. Let $\mathbf{x} = (x_1, \ldots, x_n)$ be the observed data. Past studies have suggested a Gamma(α, β) prior for μ with pre-specified values of α and β.

   (a) Find the posterior distribution of μ conditional on $\mathbf{x}$ and show that it depends on the data only through $t = \sum_{i=1}^n x_i$.

$$
\begin{aligned}
p(\mathbf{x}|\mu) &= \frac{\mu^{n\bar{x}} e^{-\mu}}{\prod_{i=1}^n x_i!} \qquad \pi(\mu) = \frac{\beta^\alpha}{\Gamma(\alpha)} \mu^{\mu-1} e^{-\beta\mu} \\
\implies p(\mu|\mathbf{x}) &\propto \mu^{n\bar{x}} \cdot e^{-\mu} \times \mu^{\alpha-1} \cdot e^{-\beta\mu} \\
&\propto \mu^{n\bar{x}+\alpha-1} \cdot e^{-\mu(\beta+1)} \\
\implies p(\mu|\mathbf{x}) &\sim \text{Gamma}(\alpha + n\bar{x}, \beta + 1)
\end{aligned}
$$

   All information of the distribution is considered know except for x, hence $p(\mu|\mathbf{x})$ depends on data through only $t = n\bar{x} = \sum_{i=1}^n x_i$.

   (b) Find the Bayes estimate for μ for squared error loss. Show that this estimate is a weighted sum of the MLE and prior (no data) estimate of μ.

$$
\begin{aligned}
L(a) &= \int_\mu (\mu - a)^2 p(\mu|\mathbf{x}) d\mu \\
\frac{\partial L(a)}{\partial a} &= -2 \int_\mu \mu p(\mu|\mathbf{x}) d\mu + 2a \int_\mu p(\mu|\mathbf{x}) d\mu \\
\implies \hat{a} &= E[\mu|\mathbf{x}] \\
&= \frac{\alpha + t}{\beta + 1} \\
&= \frac{\alpha}{\beta} \cdot \frac{\beta}{\beta+1} + t \cdot \frac{1}{\beta+1} \\
&= \frac{\alpha}{\beta} \cdot w + t \cdot (1 - w)
\end{aligned}
$$

   where $\frac{\beta}{\beta+1} = w$, is clearly a weighted sum of the MLE and prior estimate of μ.

   (c) See the following image:

(c) For arguments $y, \theta_1, \theta_2 > 0$, let function $R(y; \theta_1, \theta_2)$ denote the CDF of the gamma(θ_1, θ_2) distribution evaluated at y. Then, for arguments $p \in (0, 1)$ and $\theta_1, \theta_2 > 0$, function $R^{-1}(p; \theta_1, \theta_2)$ represents the inverse-CDF evaluated at p. In questions 1(c)i–1(c)iv below, give expressions for the following quantities using functions R and/or R^{-1}, **fully specifying all three arguments** of the functions wherever they occur in your formulas:

i. (5 points) Posterior probability, $P[\mu > \mu_0 \mid \boldsymbol{x}]$ for a known positive constant μ_0.

ii. (15 points) Lower and upper endpoints of the 95% equi-tailed Bayesian credible interval of μ.

iii. (10 points) Posterior odds, $P(H_0 \mid \boldsymbol{x})/P(H_1 \mid \boldsymbol{x})$, for testing $H_0 : \mu < 1$ versus $H_1 : \mu > 3$.

iv. (10 points) Bayes estimate of μ for a linear loss function that penalizes over- and under-estimation equally.

i.

$$P(\mu > \mu_0|\mathbf{x}) = 1 - P(\mu \leq \mu_0|\mathbf{x}) = 1 - R(\mu_0, \theta_1, \theta_2)$$

ii.

$$R^{-1}(\alpha/2, \theta_1, \theta_2) \leq \mu \leq R^{-1}(1 - \alpha/2, \theta_1, \theta_2)$$
$$R^{-1}(0.025, \theta_1, \theta_2) \leq \mu \leq R^{-1}(0.975, \theta_1, \theta_2)$$

iii.

$$\frac{P(H_0|\mathbf{y})}{P(H_1|\mathbf{y})} = \frac{P(\mu < 1|\mathbf{y})}{P(\mu > 3|\mathbf{y})} = \frac{R(1, \theta_1, \theta_2)}{1 - R(3, \theta_1, \theta_2)}$$

iv.

$$R(\hat{a}, \theta_1, \theta_2) = \frac{c_2}{c_1 + c_2} \overbrace{=}^{\text{over-under equal}} \frac{1}{2}$$
$$\implies \hat{a} = R^{-1}(1/2, \theta_1, \theta_2)$$

(d) Before if before data collections, previous works assumed $E[\mu] = m$, and $\sqrt{\text{var}(\mu)} = s$, what would you recommend as the specified prior values for α and β?
Since $\pi(\mu) \sim \text{Gamma}(\alpha, \beta)$, we can say that this is equiv to saying

$$E[\pi(\mu)] = \frac{\alpha}{\beta} = m \qquad \text{var}[\pi(\mu)] = \frac{\alpha}{\beta^2} = s^2$$
$$\implies \alpha = \frac{m^2}{s^2} \qquad \beta = \frac{m}{s^2}$$

2. Suppose that the model is instead $X_1, \ldots, X_n \sim_{i.i.d.} \text{NegBin}(\mu, b)$ for unknown parameter μ and known parameter b. Assume the prior is the same. Describe the rejection algorithm for sampling the posterior to the point where it could be easily coded in R.
The MLE of μ is defined as

$$\mu_{MLE} = \bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i$$

where the x_i's are from the given sample. The likelihood function for the negative binomial data is

$$p(\mathbf{x}|\mu, b) = \left[\prod_{i=1}^{n} \frac{\Gamma(x_i + b)}{\Gamma(x_i + 1)\Gamma(b)}\right] \left(\frac{b}{\mu + b}\right)^{nb} \left(\frac{\mu}{\mu + b}\right)^{n\bar{x}}$$

Let M be the value of the likelihood function at the MLE.

$$M = p(x|\bar{x}, b) = \left[\prod_{i=1}^{n} \frac{\Gamma(x_i + b)}{\Gamma(x_i + 1)\Gamma(b)}\right] \left(\frac{b}{\bar{x} + b}\right)^{nb} \left(\frac{\bar{x}}{\bar{x} + b}\right)^{n\bar{x}}$$

Consider then the fraction

$$\frac{p(x|\mu, b)}{p(x|\bar{x}, b)} = \left(\frac{\mu(\bar{x} + b)}{\bar{x}(\mu + b)}\right)^{n\bar{x}} \left(\frac{\bar{x} + b}{\mu + b}\right)^{nb} = f(\mu|b, \mathbf{x})$$

as a function of μ. Then, the algorithm is as follows:

(a) Generate $T \sim \mathrm{U}(0, 1)$, and independently $\hat{\mu} \sim \text{Gamma}(\alpha, \beta)$.

(b) Check if

$$T < f(\hat{\mu}|b, \mathbf{x})$$

is true, if it is keep $\hat{\mu}$ as a valid sample, otherwise return to 1.

The values of $n\bar{x}$ in $f(\hat{\mu}|b, \mathbf{x})$ come from the sample used to establish the negative binomial model of the likelihood.

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